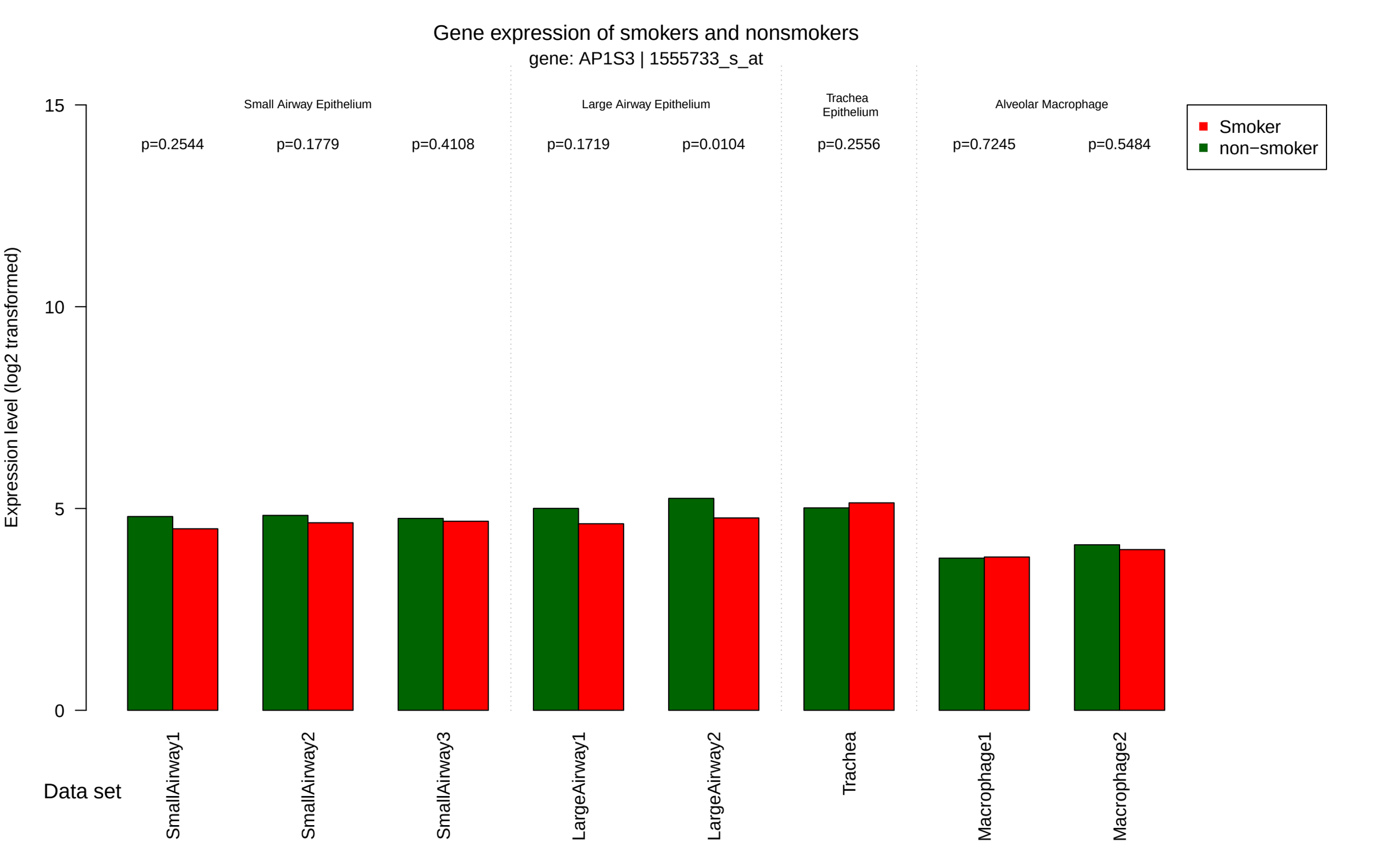
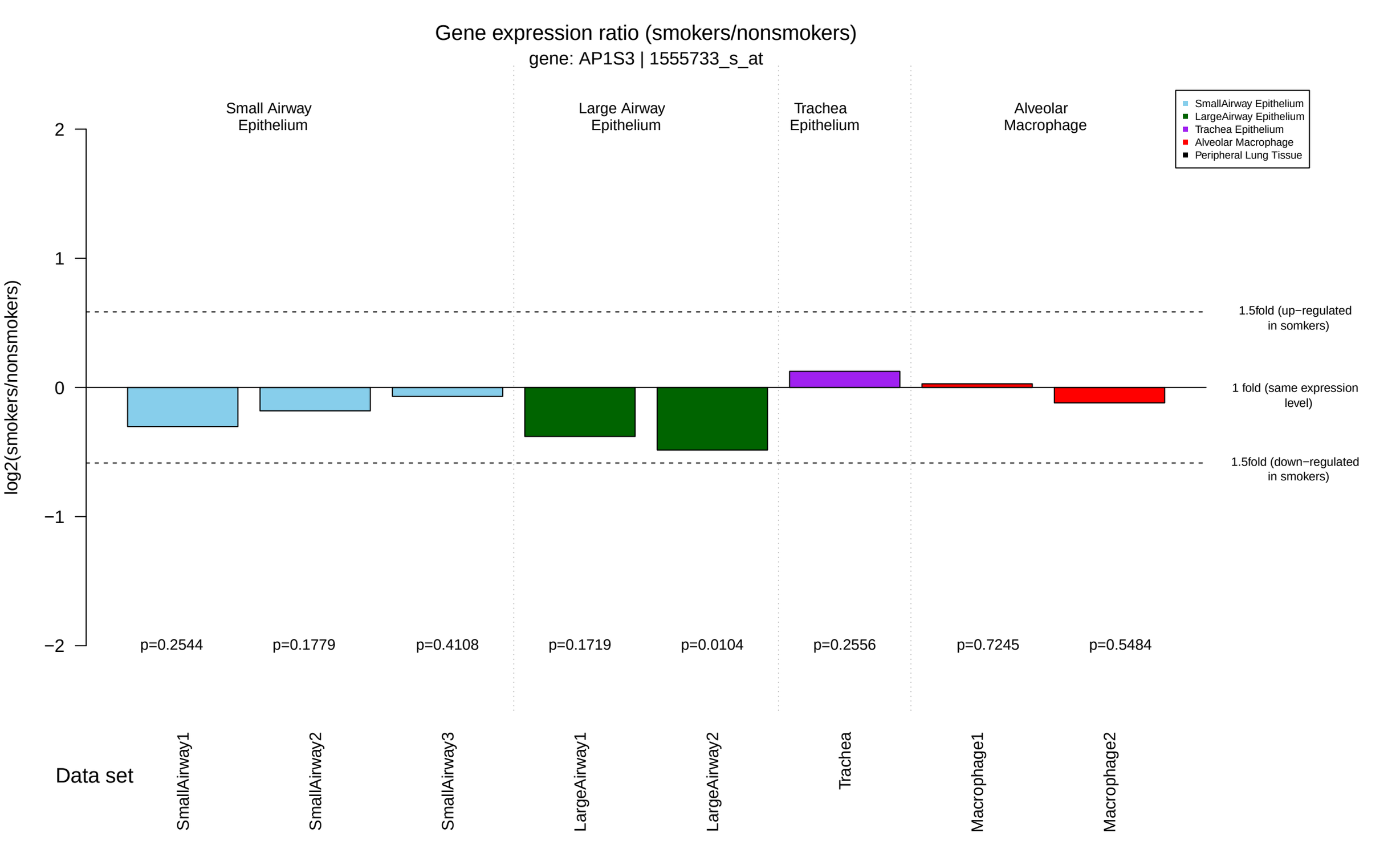
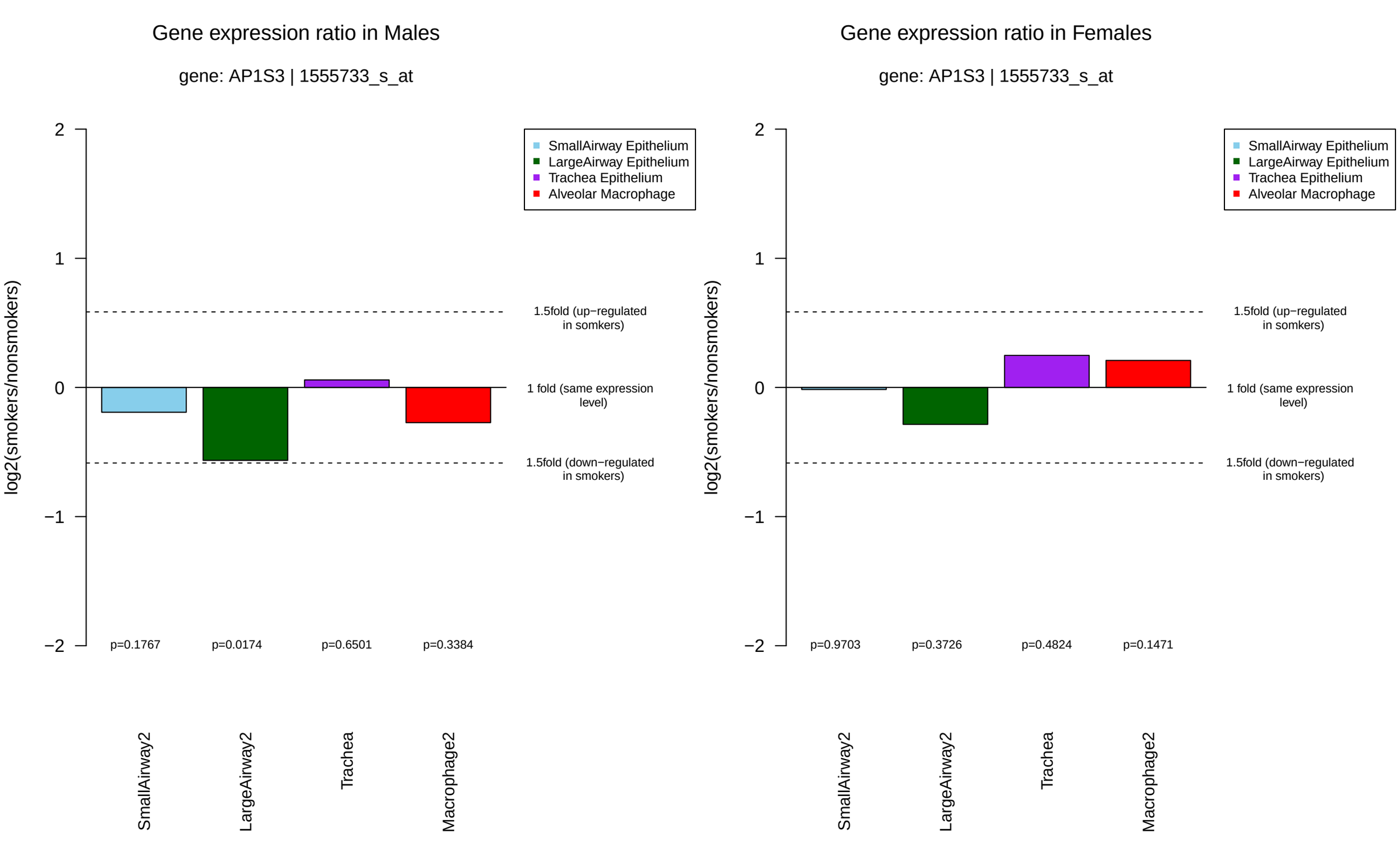
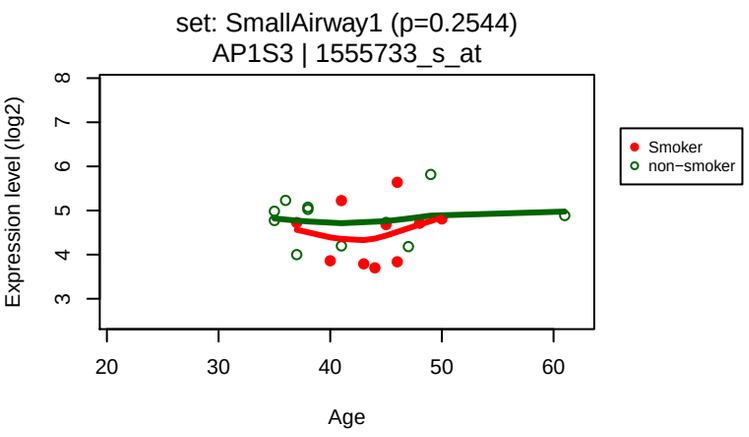
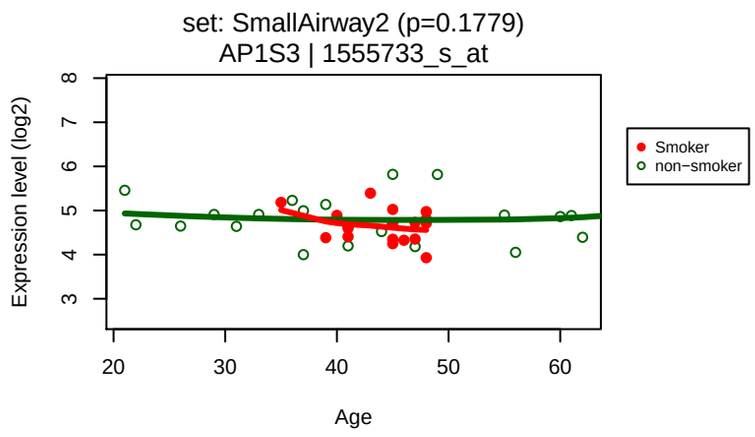
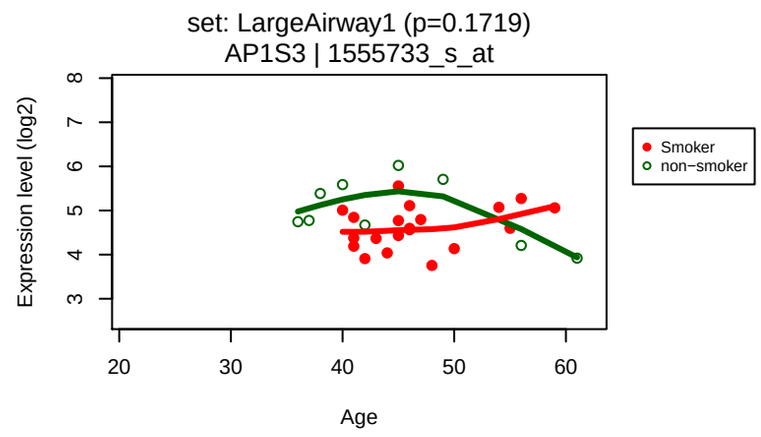

Gene expression of smokers and nonsmokers
gene: AP1S3 | 1555733_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2544
p=0.1779
p=0.4108
p=0.1719
p=0.0104
p=0.2556
p=0.7245
p=0.5484
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

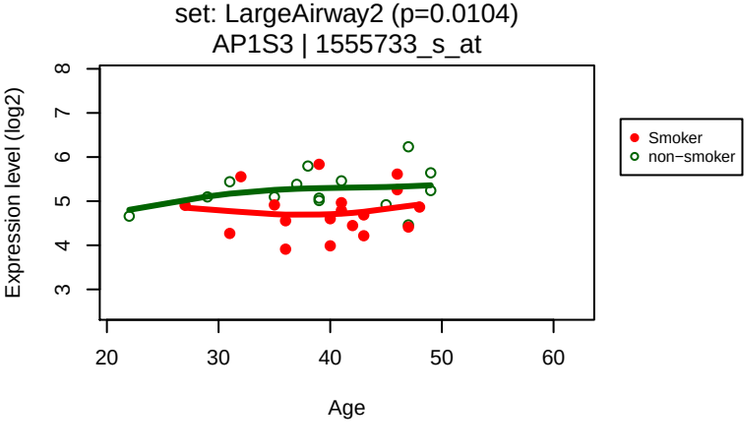

Gene expression ratio (smokers/nonsmokers)
gene: AP1S3 | 1555733_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2544
p=0.1779
p=0.4108
p=0.1719
p=0.0104
p=0.2556
p=0.7245
p=0.5484
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

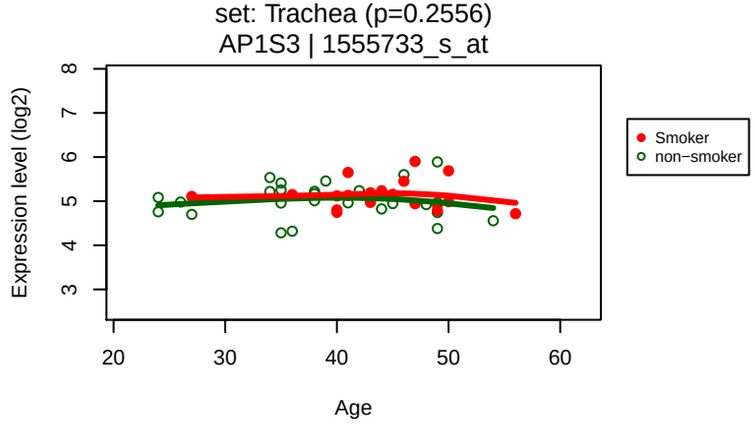

Gene expression ratio in Males
gene: AP1S3 | 1555733_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1767
p=0.0174
p=0.6501
p=0.3384
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AP1S3 | 1555733_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9703
p=0.3726
p=0.4824
p=0.1471
SmallAirway2
LargeAirway2
Trachea
Macrophage2

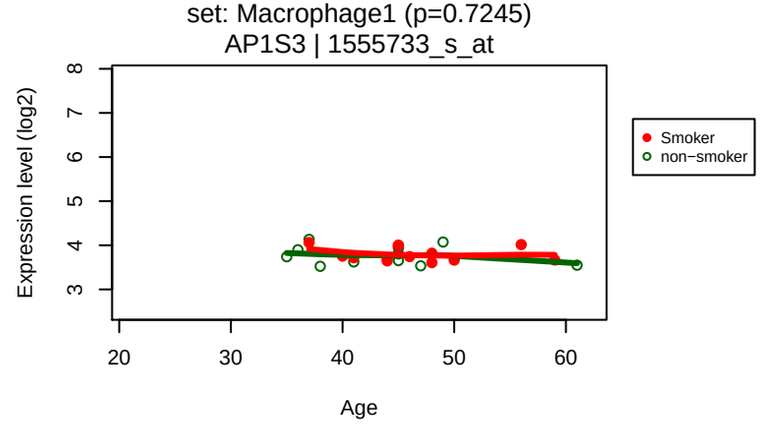

set: SmallAirway1 (p=0.2544)
AP1S3 | 1555733_s_at
Expression level (log2)
Age
Smoker
non−smoker

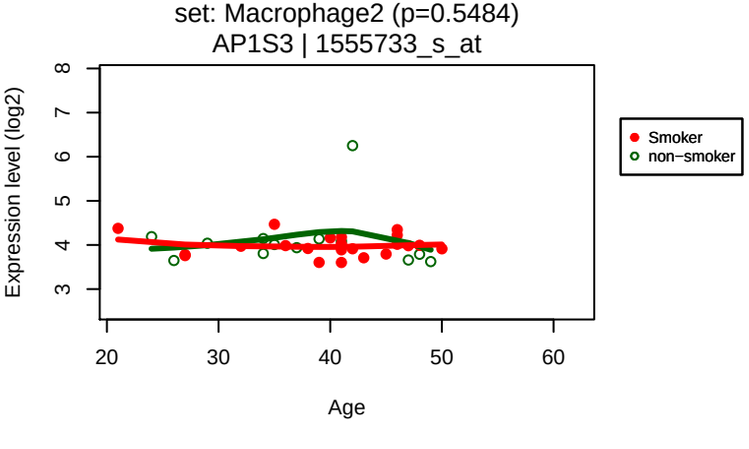

set: SmallAirway2 (p=0.1779)
AP1S3 | 1555733_s_at
Expression level (log2)
Age
Smoker
non−smoker

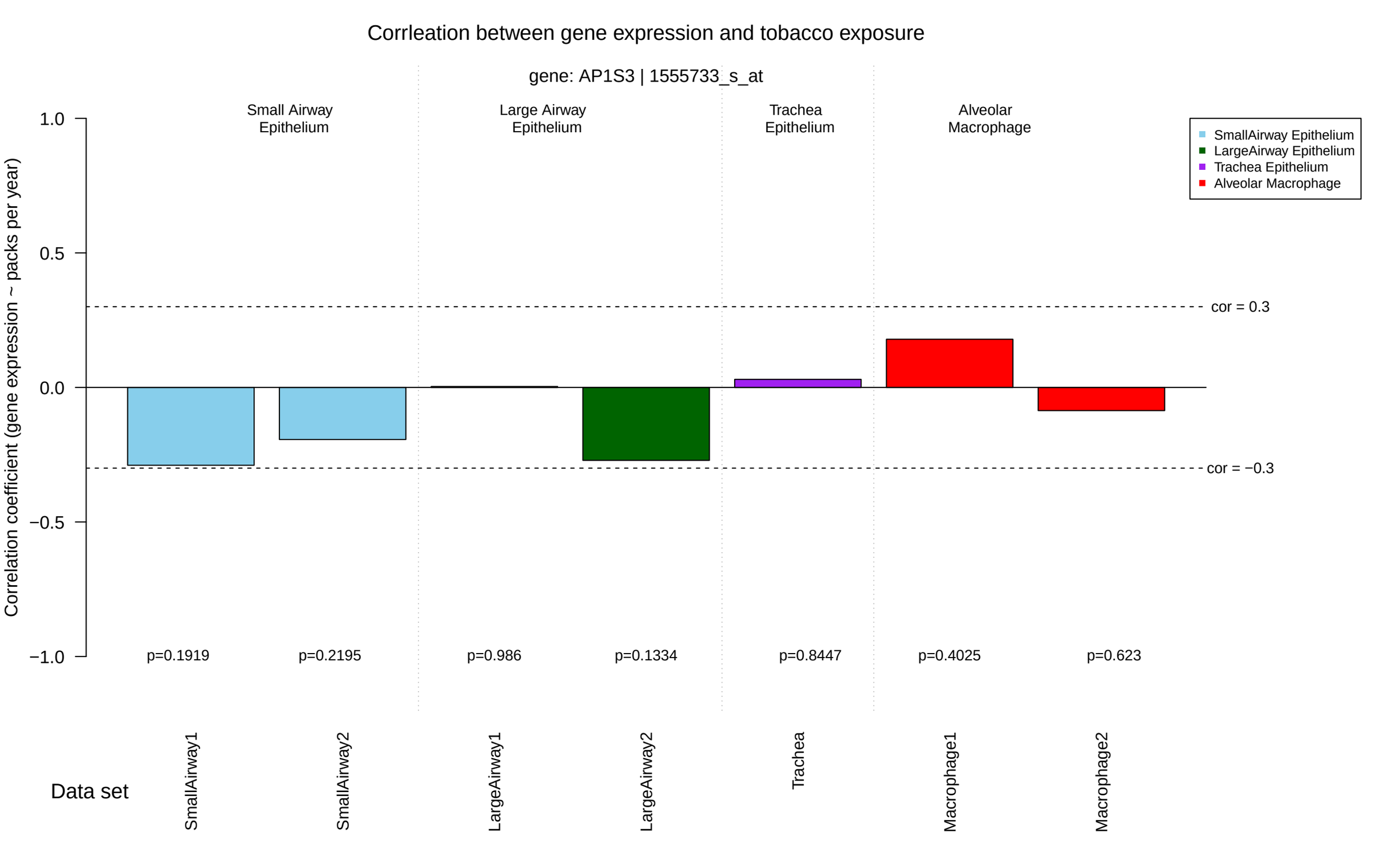

set: LargeAirway1 (p=0.1719)
AP1S3 | 1555733_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0104)
AP1S3 | 1555733_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2556)
AP1S3 | 1555733_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7245)
AP1S3 | 1555733_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5484)
AP1S3 | 1555733_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AP1S3 | 1555733_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1919
p=0.2195
p=0.986
p=0.1334
p=0.8447
p=0.4025
p=0.623
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2